

Ho Ko

AI/ML SOFTWARE ENGINEER

CONTACT

EMAIL

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LOCATION

San Francisco Bay Area, CA

WEBSITE

hoko.xyz

LINKEDIN

linkedin.com/in/hoko

GITHUB

github.com/woodyhoko

SKILLS

Languages

- Python
- C/C++
- Swift
- Java
- Kotlin
- JavaScript
- x86 Assembly
- MATLAB

ML Lifecycle

- Model Training
- Distillation
- Quantization & Compression
- Inference Engines
- On-Device Deployment
- Scalable Serving

On-Device & Web

- WebGPU
- Three.js
- Transformers.js
- MediaPipe
- WebRTC
- Local LLMs (Gemma, LFM)

Domains

- NLP
- Computer Vision
- Healthcare
- Astronomy
- Weather Forecasting
- Economics

EDUCATION

M.S. Computer Science

USC — AI & ML

2021–2023 · GPA 4.0/4.0

M.S. Electrical & Computer Eng.

Waterloo — ML & AI

2020–2021 · GPA 3.9/4.0

B.S. Computer Science

NCU, Taiwan

2016–2020 · GPA 3.8/4.0

AI/ML Software Engineer with cross-domain experience spanning astronomy, weather forecasting, economics, healthcare, NLP, and computer vision. Works across the full **model lifecycle** — training and distilling deep models, architecting highly efficient inference engines, and building scalable, high-throughput serving — and brings that intelligence **on-device** through WebGPU, Three.js, and local LLMs in privacy-preserving systems that have reached **1B+ users**.

EXPERIENCE

Machine Learning Engineer · Google

2026–Present

- Build **knowledge-grounded retrieval & ranking models** on Google Lens that power AI Mode in Google Search and AI Overviews at Search scale.
- Distill Gemini** into highly compressed models that hold near-equal quality while meeting the Lens stack's latency, scalability, and multimodal requirements.

Software Engineer · Google

2023–2026

- Core contributor to the **LiteRT LM inference engine**, running LLMs privately and offline across Windows, macOS, Linux, and Android — including NPU acceleration.
- Enabled Gemma multimodality (vision + audio + text) and Gemini Nano on-device; helped ship **Chrome Built-in AI and Google AI Core, reaching 1B+ users**.
- Shipped the **Google AI Edge Gallery** app on Android and iOS.

Research Collaborator · Google DeepMind

2024–2025

- Designed **Sparse RAG (ICLR 2025)**, a sparse decoding paradigm that encodes retrieved docs in parallel and attends only to the most relevant caches via learned control tokens.
- Cut long-range attention latency while **improving generation quality across 4 retrieval benchmarks**.

Research Collaborator · Google Health & Fitbit

2024–2025

- Built and evaluated ML models **detecting hypertension risk from raw wrist PPG** passively collected on a consumer smartwatch (medRxiv), enabling non-invasive screening at population scale.

Software Engineer · MyCareLinq

2023

- Built **NLP pipelines and a home-health-care knowledge graph** from unstructured health documentation.
- Integrated LLMs powering caregiving assistants that surface contextual recommendations and care-plan summaries.

Software Engineer Intern · Google

2022

- Built a TFLite tensor extractor + metadata-writer with classification rules at **100% coverage of common GenAI task libraries** (LiteRT Task Library).
- Integrated a **C++ code generator** compiling optimized inference pipelines for arbitrary TFLite files.

Research Assistant · Caltech (SURF)

2018

- Analyzed transient imaging from the Zwicky Transient Facility; **co-authored a Type IIb supernova study in ApJL (878:L5)** and simulated double-peak light-curve models.

SELECTED PUBLICATIONS

- Accelerating Inference of Retrieval-Augmented Generation via Sparse Context Selection — ICLR 2025**, Google DeepMind.
- Opportunistic Hypertension Detection from Wearable PPG — medRxiv 2025**, Google Health & Fitbit.
- A Type IIb Supernova (ZTF18aalrxas) — ApJL 878:L5**, 2019.